3 Vectors and Directions

Problem 3.1: Vector Directions

Let $\vec{v} = \langle 3, -4, 2 \rangle$. Find a unit vector in the same direction as $\vec{v}$.

Solution 3.1

To find a unit vector in the same direction as $\vec{v}$, we first compute its magnitude:

$$\|\vec{v}\| = \sqrt{3^2 + (-4)^2 + 2^2} = \sqrt{9 + 16 + 4} = \sqrt{29}$$

The unit vector in the same direction as $\vec{v}$ is then:

$$\hat{v} = \frac{\vec{v}}{\|\vec{v}\|} = \frac{1}{\sqrt{29}} \langle 3, -4, 2 \rangle$$

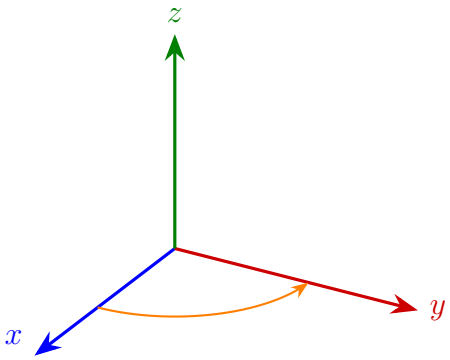
In general, for any non-zero vector $\vec{w}$,

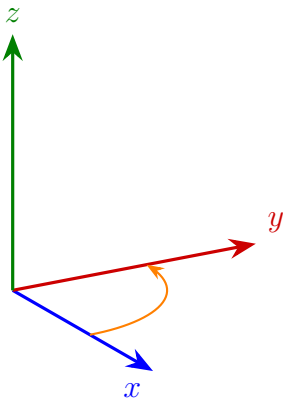
$$\begin{aligned} \|\lambda\vec{w}\| &= \sqrt{(\lambda w_1)^2 + (\lambda w_2)^2 + (\lambda w_3)^2} \\ &= \sqrt{\lambda^2(w_1^2 + w_2^2 + w_3^2)} \\ &= |\lambda|\|\vec{w}\| \end{aligned}$$

Hence, if we want a unit vector in the same direction as $\vec{w}$, we can choose $\lambda = \dfrac{1}{\|\vec{w}\|}$, giving us:

$$\hat{w} = \frac{\vec{w}}{\|\vec{w}\|}$$

Right-Hand Rule The relative position of the x-, y-, and z-axes in $\mathbb{R}^3$ is determined by the *right-hand rule*. If you point your right thumb along the positive x-axis and your index finger along the positive y-axis, then your middle finger (perpendicular to both) will point along the positive z-axis.





Definition 3.1: Parameterizations

A parameterization is a function whose range is a given geometric object.

For example, a parameterization of the unit circle is given by:

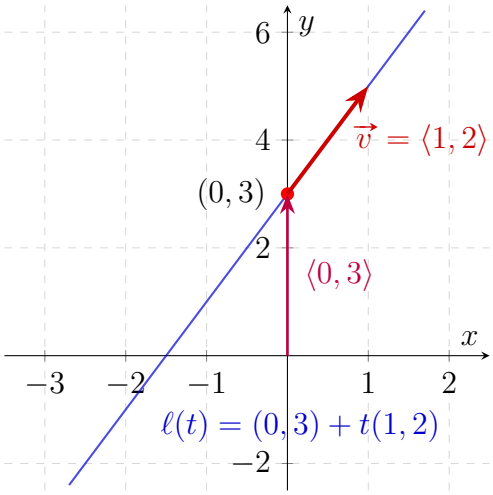
$$f(t) = (\cos t, \sin t)$$

Example 3.1.

To parameterize the line $y = 2x + 3$:

$$\begin{aligned} \ell(t) &= (t, 2t + 3) \\ &= \langle 0, 3\rangle + \langle t, 2t\rangle \end{aligned}$$

The plot is then given:



To generalize, to parameterize a line in $\mathbb{R}^2$, we can first take a point on the line. Then, we

take some $\vec{v}$ in the direction of the line. Then, we can write the parameterization as:

$$\ell(t) = (p_1 + v_1 t,\ p_2 + v_2 t)$$

In $\mathbb{R}^3$, we can do the same thing.

Example 3.2.

Parameterize the line passing through $P(1, 2, 0)$ and $Q(5, -1, 1)$.

$$\begin{aligned}\overrightarrow{PQ} &= \langle 5-1, -1-2, 1-0\rangle \\ &= \langle 4, -3, 1\rangle \\ \ell(t) &= (1, 2, 0) + t(4, -3, 1) \\ &= (1 + 4t, 2 - 3t, t)\end{aligned}$$

3.1 Dot Product

Definition 3.2: Dot Product

The dot product of two vectors $\vec{v} = \langle v_1, v_2\rangle$ and $\vec{w} = \langle w_1, w_2\rangle$ is defined as:

$$\vec{v} \cdot \vec{w} = v_1 w_1 + v_2 w_2$$

Example 3.3.

$$\vec{v} \cdot \vec{v} = v_1^2 + v_2^2 = \|\vec{v}\|^2$$
$$\vec{i} \cdot \vec{j} = 1 \cdot 0 + 0 \cdot 1 = 0$$

Problem 3.2: Dot Product and Angle

Find pairs of vectors with:

- Dot product > 0
- Dot product $= 0$

- Dot product < 0

Solution 3.2

- $\vec{v} \cdot \vec{v} = \|\vec{v}\|^2 > 0$ for any non-zero vector $\vec{v}$.
- $\vec{i} \cdot \vec{j} = 0$ since they are orthogonal.
- $\langle -1, -1 \rangle \cdot \langle 1, 1 \rangle = -1 - 1 = -2 < 0$.

To conclude, the dot products of any two vectors corresponds to the following situations:

- Dot product $= 0$ if and only if the two vectors are orthogonal (perpendicular).
- Dot product > 0 if the angle between the two vectors is acute (less than 90 degrees).
- Dot product < 0 if the angle between the two vectors is obtuse (greater than 90 degrees).

This is because $\vec{v} \cdot \vec{w} = \|\vec{v}\|\|\vec{w}\| \cos\theta$, where θ is the angle between the vectors. Hence,

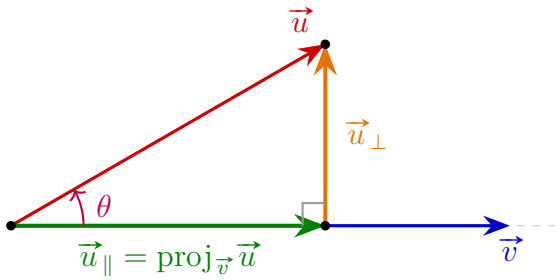
$$\theta = \arccos\left(\frac{\vec{v} \cdot \vec{w}}{\|\vec{v}\|\|\vec{w}\|}\right)$$

Definition 3.3: Orthogonal Projection

Write $\vec{u} = \vec{u}_{\parallel} + \vec{u}_{\perp}$, where $\vec{u}_{\parallel} \parallel \vec{v}$ and $\vec{u}_{\perp} \perp \vec{u}_{\parallel}$. Then, $\vec{u}_{\parallel}$ is called the **orthogonal projection** of $\vec{u}$ onto $\vec{v}$.

Example 3.4.

Below is an illustration of orthogonal projection:



By this, we could see that

$$\vec{u} = \vec{u}_{\parallel} + \vec{u}_{\perp}$$

Then, we solve for $\vec{u}_{\parallel}$ and $\vec{u}_{\perp}$.

Taking the dot product with $\vec{v}$:

$$\begin{aligned}
\vec{u} \cdot \vec{v} &= (\vec{u}_{\parallel} + \vec{u}_{\perp}) \cdot \vec{v} \\
&= \vec{u}_{\parallel} \cdot \vec{v} + \vec{u}_{\perp} \cdot \vec{v} \\
&= \vec{u}_{\parallel} \cdot \vec{v} + 0 \quad (\text{since } \vec{u}_{\perp} \perp \vec{v}) \\
&= \vec{u}_{\parallel} \cdot \vec{v}
\end{aligned}$$

Let $\vec{e}_v = \dfrac{1}{\|\vec{v}\|}\vec{v}$ be the unit vector in the direction of $\vec{v}$, and write $\vec{u}_{\parallel} = \lambda \vec{e}_v$. Then:

$$\begin{aligned}
\vec{u} \cdot \vec{v} &= (\lambda \vec{e}_v) \cdot (\|\vec{v}\| \vec{e}_v) \\
&= \lambda \|\vec{v}\| (\vec{e}_v \cdot \vec{e}_v) \\
&= \lambda \|\vec{v}\| \|\vec{e}_v\|^2 \\
&= \lambda \|\vec{v}\| \quad (\text{since } \|\vec{e}_v\| = 1)
\end{aligned}$$

Solving for λ:

$$\lambda = \frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|}$$

Thus, the parallel projection vector $\vec{u}_{\parallel}$ and the orthogonal component $\vec{u}_{\perp}$ are given by:

$$\begin{aligned}
\vec{u}_{\parallel} &= \left(\frac{\vec{u}\cdot\vec{v}}{\|\vec{v}\|}\right)\vec{e}_{v} \\
&= \frac{\vec{u}\cdot\vec{v}}{\|\vec{v}\|^{2}}\vec{v} \\
&= \frac{\vec{u}\cdot\vec{v}}{\vec{v}\cdot\vec{v}}\vec{v} \\
\vec{u}_{\perp} &= \vec{u} - \vec{u}_{\parallel} \\
&= \vec{u} - \frac{\vec{u}\cdot\vec{v}}{\vec{v}\cdot\vec{v}}\vec{v}
\end{aligned}$$

Furthermore, the magnitude of the parallel projection is:

$$\left\|\vec{u}_{\parallel}\right\| = \|\vec{u}\|\,|\cos\theta|$$

Problem 3.3: Vector Projection

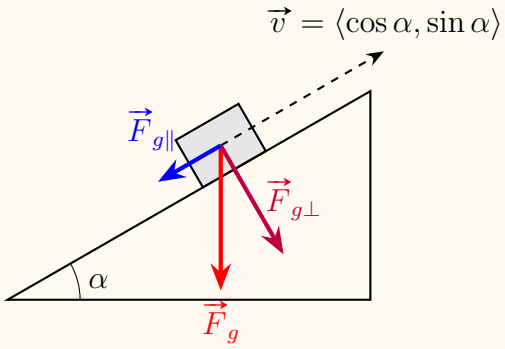
Find the projection of $\vec{u} = \langle 3, 2, 2\rangle$ onto $\vec{v}\,\langle 1, -2, 0\rangle$

Solution 3.3

$$\begin{aligned}
\vec{u}_{\parallel} &= \frac{3\cdot 1 + 2\cdot(-1) + 2\cdot 0}{1^{2} + (-1)^{2} + 0^{2}}\langle 1, -1, 0\rangle \\
&= \frac{1}{2}\langle 1, -1, 0\rangle \\
&= \left\langle \frac{1}{2}, -\frac{1}{2}, 0 \right\rangle
\end{aligned}$$

Problem 3.4: Acceleration on an Inclined Plane

Consider a block sliding down an inclined plane with an angle of inclination α.



Solution 3.4

To find the magnitude of acceleration down the incline, we project the gravitational force $\vec{F}_g$ along the incline direction:

$$\begin{aligned}
\text{acceleration} &= \left\| \vec{F}_{g\parallel} \right\| \\
&= \left\| \vec{F}_g \right\| |\cos\theta| \\
&= \left\| \vec{F}_g \right\| \sin\alpha = 9.8 \sin\alpha \quad (\text{m/s}^2)
\end{aligned}$$

Problem 3.5: Vector Projection

Find the projection of $\vec{u} = \langle 3, 2, 2 \rangle$ onto $\vec{v} = \langle 1, -1, 0 \rangle$.

Solution 3.5

Using the vector projection formula:

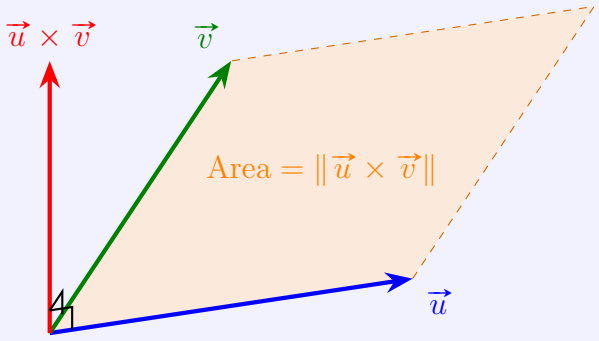
$$\begin{aligned}
\vec{u}_{\parallel} &= \frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|^2} \vec{v} \\
&= \frac{3(1) + 2(-1) + 2(0)}{1^2 + (-1)^2 + 0^2} \langle 1, -1, 0 \rangle \\
&= \frac{3 - 2 + 0}{1 + 1 + 0} \langle 1, -1, 0 \rangle \\
&= \frac{1}{2} \langle 1, -1, 0 \rangle \\
&= \left\langle \frac{1}{2}, -\frac{1}{2}, 0 \right\rangle
\end{aligned}$$

3.2 Determinants and Cross Products

Definition 3.4: Vector Cross Product

Let $\vec{u}, \vec{v} \in \mathbb{R}^3$. The **cross product** $\vec{u} \times \vec{v}$ is a vector in $\mathbb{R}^3$ satisfying the following three geometric properties:

1. **Orthogonality:** $\vec{u} \times \vec{v} \perp \vec{u}$ and $\vec{u} \times \vec{v} \perp \vec{v}$.
2. **Magnitude:** $\|\vec{u} \times \vec{v}\| = \|\vec{u}\|\|\vec{v}\| \sin\theta$, which equals the area of the parallelogram formed by $\vec{u}$ and $\vec{v}$.
3. **Orientation:** The ordered triple $(\vec{u}, \vec{v}, \vec{u} \times \vec{v})$ is right-handed (positively oriented).



From Property (2), we have for the standard basis vectors:

$$\left\|\vec{i} \times \vec{i}\right\| = \left\|\vec{i}\right\|\left\|\vec{i}\right\| \sin 0^\circ = 0 \implies \vec{i} \times \vec{i} = \vec{0}$$

And by the Right-Hand Rule:

$$\vec{i} \times \vec{j} = \vec{k}$$

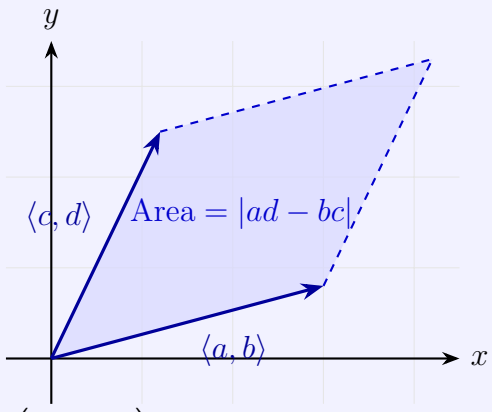
Definition 3.5: Determinants

For a 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the determinant is defined as:

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

This is denoted as $\det(A)$ or $|A|$ for a 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

For a 2×2 matrix, the determinant represents the area of the parallelogram spanned by $\langle a, c \rangle$ and $\langle b, d \rangle$ (the column vectors) or $\langle a, b \rangle$ and $\langle c, d \rangle$.

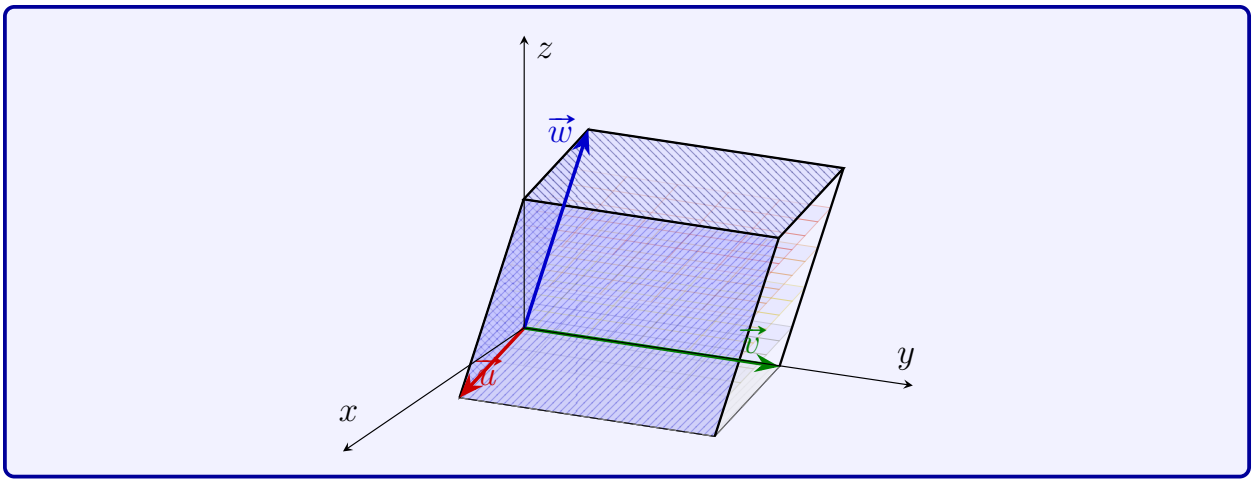


For a 3×3 matrix $B = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$, the determinant is defined as:

$$\det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

$$= aei + bfg + cdh - ceg - bdi - afh$$

$\det \begin{pmatrix} \vec{u} \\ \vec{v} \\ \vec{w} \end{pmatrix}$ represents the volume of the parallelepiped spanned by $\vec{u}$, $\vec{v}$, and $\vec{w}$.



The cross product of two vectors $\vec{u} = \langle u_1, u_2, u_3 \rangle$ and $\vec{v} = \langle v_1, v_2, v_3 \rangle$ can be computed using the determinant of a 3×3 matrix:

$$\vec{u} \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = \vec{i} \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} - \vec{j} \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} + \vec{k} \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix}$$

Problem 3.6: Cross Product

Let $\vec{v} = \langle 3, 2, -1 \rangle$, $\vec{w} = \langle 4, 1, -3 \rangle$. Find $\vec{v} \times \vec{w}$.

Solution 3.6

$$
\begin{aligned}
\vec{v} \times \vec{w} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & 2 & -1 \\ 4 & 1 & -3 \end{vmatrix} \\
&= \vec{i} \begin{vmatrix} 2 & -1 \\ 1 & -3 \end{vmatrix} - \vec{j} \begin{vmatrix} 3 & -1 \\ 4 & -3 \end{vmatrix} + \vec{k} \begin{vmatrix} 3 & 2 \\ 4 & 1 \end{vmatrix} \\
&= \vec{i}\,(2(-3) - (-1)(1)) - \vec{j}\,(3(-3) - (-1)(4)) + \vec{k}\,(3(1) - 2(4)) \\
&= \vec{i}\,(-6+1) - \vec{j}\,(-9+4) + \vec{k}\,(3-8) \\
&= \vec{i}\,(-5) - \vec{j}\,(-5) + \vec{k}\,(-5) \\
&= -5\vec{i} + 5\vec{j} - 5\vec{k}
\end{aligned}
$$

3.2.1 Properties of the Cross Product

- $\vec{v} \times \vec{w} \perp \vec{v}$ and $\vec{v} \times \vec{w} \perp \vec{w}$. This implies that $\vec{v} \cdot (\vec{v} \times \vec{w}) = 0$ and $\vec{w} \cdot (\vec{v} \times \vec{w}) = 0$.
- $\{\vec{v}, \vec{w}, \vec{v} \times \vec{w}\}$ forms a right-handed system.
- $\|\vec{v} \times \vec{w}\|$ represents the area of the parallelogram spanned by $\vec{v}$ and $\vec{w}$.

3.2.2 Algebraic Rules of the Cross Product

- Anti-commutativity: $\vec{v} \times \vec{w} = -(\vec{w} \times \vec{v})$
- Parallel vectors: $\vec{v} \times \vec{w} = \vec{0} \iff \vec{v} = \lambda\vec{w}$ or $\vec{w} = \vec{0}$.
- Scalar multiplication: $\lambda(\vec{v} \times \vec{w}) = (\lambda\vec{v}) \times \vec{w} = \vec{v} \times (\lambda\vec{w})$
- Distributivity: $(\vec{u} + \vec{v}) \times \vec{w} = \vec{u} \times \vec{w} + \vec{v} \times \vec{w}$
- Standard basis relations:

$$\vec{i} \times \vec{j} = \vec{k}, \quad \vec{j} \times \vec{k} = \vec{i}, \quad \vec{k} \times \vec{i} = \vec{j}$$

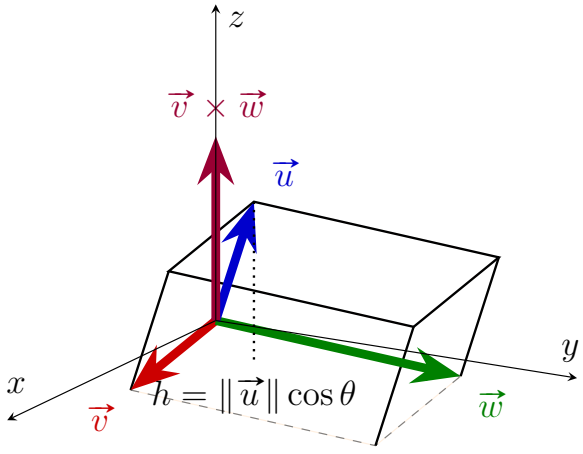
Hence, we may expand the cross product of two vectors $\vec{v} = \langle v_1, v_2, v_3 \rangle$ and $\vec{w} = \langle w_1, w_2, w_3 \rangle$ component-wise:

$$\begin{aligned}
\vec{v} \times \vec{w} &= (v_1\,\vec{i} + v_2\,\vec{j} + v_3\,\vec{k}) \times (w_1\,\vec{i} + w_2\,\vec{j} + w_3\,\vec{k}) \\
&= v_1w_1(\vec{i} \times \vec{i}) + v_1w_2(\vec{i} \times \vec{j}) + v_1w_3(\vec{i} \times \vec{k}) \\
&\quad + v_2w_1(\vec{j} \times \vec{i}) + v_2w_2(\vec{j} \times \vec{j}) + v_2w_3(\vec{j} \times \vec{k}) \\
&\quad + v_3w_1(\vec{k} \times \vec{i}) + v_3w_2(\vec{k} \times \vec{j}) + v_3w_3(\vec{k} \times \vec{k}) \\
&= v_1w_2\,\vec{k} - v_1w_3\,\vec{j} - v_2w_1\,\vec{k} + v_2w_3\,\vec{i} + v_3w_1\,\vec{j} - v_3w_2\,\vec{i} \\
&= (v_2w_3 - v_3w_2)\,\vec{i} + (v_3w_1 - v_1w_3)\,\vec{j} + (v_1w_2 - v_2w_1)\,\vec{k}
\end{aligned}$$

3.3 Scalar Triple Product

Given $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^3$, the quantity $\vec{u} \cdot (\vec{v} \times \vec{w})$ is defined as the **scalar triple product**.

The absolute value $|\vec{u} \cdot (\vec{v} \times \vec{w})|$ gives the geometric volume of the parallelepiped spanned by $\vec{u}$, $\vec{v}$, and $\vec{w}$:



The sign of $\vec{u} \cdot (\vec{v} \times \vec{w})$ indicates the orientation of the vector basis.

$\vec{u} \cdot (\vec{v} \times \vec{w}) > 0$ if the ordered triple $(\vec{u}, \vec{v}, \vec{w})$ forms a **right-handed system** (i.e., $\vec{u}$ points to the same side of the plane spanned by $\vec{v}$ and $\vec{w}$ as $\vec{v} \times \vec{w}$).